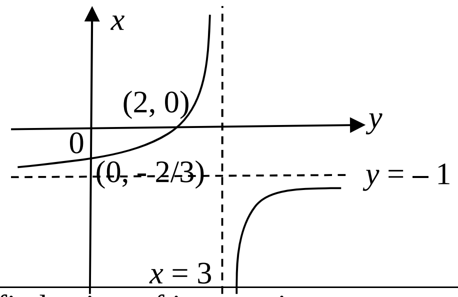
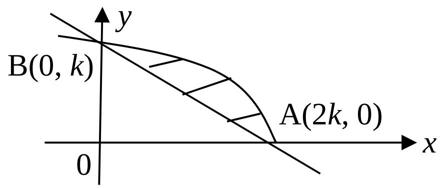
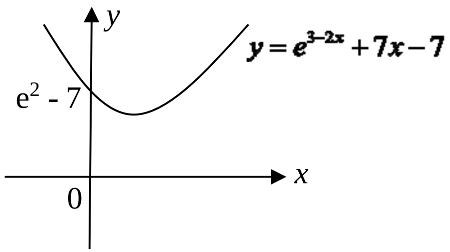
	Suggested Solutions
1i	When $x = 0$, $y = -\frac{2}{3}$ When $y = 0$, $x = 2$ V.A. is $x = 3$, H.A. , using long division, $y = \frac{2-x}{x-3} = -\frac{x-2}{x-3} = -\frac{x-3+1}{x-3} = -1 - \frac{1}{x-3}$ So, H.A. is $y = -1$
	
1ii	To find points of intersection, $\frac{2-x}{x-3} = kx - 1$, for $x \neq 3$
	$2 - x = (kx-1)(x-3)$ $2 - x = kx^2 - x - 3kx + 3$ $kx^2 - 3kx + 1 = 0$ For two distinct points of intersection, $b^2 - 4ac > 0$
	$(-3k)^2 - 4(k)(1) > 0$
	$k(9k-4) > 0$  So $k < 0$ or $k > 4/9$ Since $k > 0$, hence $\{k \in \mathbb{R}: k > 4/9\}$
2i	volume of box, $V = (2x)(x)(y)$

	$2x^2y = 24\,000$ $y = \dfrac{12\,000}{x^2}$
	$P = 4x + 4(2x) + 4y$ $= 12x + 4(\dfrac{12\,000}{x^2})$
	$= 12x + \dfrac{48\,000}{x^2}$ (Shown)
2ii	$\dfrac{dP}{dx} = 12 - 2(48\,000\, x^{-3})$
	For stationary values, set $\dfrac{dP}{dx} = 0$ $12 - 2(48\,000\, x^{-3}) = 0$
	$12 - \dfrac{2 \times 48\,000}{x^3} = 0$ $\dfrac{1}{x^3} = \dfrac{12}{2 \times 48\,000} = \dfrac{1}{8000}$ $x^3 = 8000$ $x = 20$ cm
	$\begin{array}{c\|c\|c\|c} x & 20^- & 20 & 20^+ \\ \hline \dfrac{dP}{dx} & -\text{ ve} & 0 & +\text{ve} \end{array}$
	So $x = 20$ cm gives P min Minimum value of $P = 12(20) + \dfrac{48\,000}{(20)^2}$ $= 240 + 120 = 360$ cm.
3(a)	$\dfrac{d}{dx} \ln\left(\dfrac{20x}{10 - x}\right)$ $= \dfrac{d}{dx}[\ln 20x - \ln(10 - x)]$

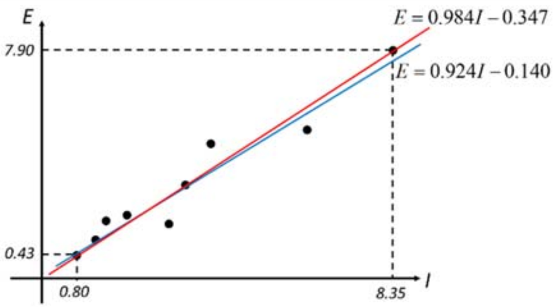
	$=\dfrac{1}{x}+\dfrac{1}{10-x}$
3(b)	$\dfrac{d}{dx}\left(\dfrac{1}{1+x^2}\right)=-\dfrac{1}{(1+x^2)^2}(2x)=-\dfrac{2x}{(1+x^2)^2}$
	$\int_0^2 \dfrac{x}{(1+x^2)^2}\,dx$ $=-\dfrac{1}{2}\int_0^2 \dfrac{-2x}{(1+x^2)^2}\,dx$
	$=-\dfrac{1}{2}\left[\dfrac{1}{1+x^2}\right]_0^2$ $=-\dfrac{1}{2}\left(\dfrac{1}{5}-1\right)$ $=-\dfrac{1}{2}\left(-\dfrac{4}{5}\right)$ $=\dfrac{2}{5}$
3(c)	$\int_1^a \left(e^{1-x}+\dfrac{1}{x^2}\right)dx=-\dfrac{1}{a}$ $\left[-e^{1-x}-\dfrac{1}{x}\right]_1^a=-\dfrac{1}{a}$
	$-e^{1-a}-\dfrac{1}{a}-\left(-e^{1-1}-\dfrac{1}{1}\right)=-\dfrac{1}{a}$ $-e^{1-a}-\dfrac{1}{a}+1+1=-\dfrac{1}{a}$ $-e^{1-a}=-2$ $1-a=\ln 2$ $a=1-\ln 2$ $-e^{1-a}=-2$ $1-a=\ln 2$ $a=1-\ln 2$
4i	Since the line *AB* passes through points $A(2k, 0)$ and $B(0, k)$, Gradient of the line *AB* $=\dfrac{k-0}{0-2k}=-\dfrac{1}{2}$

	Hence equation of line AB is $\dfrac{y-0}{x-2k}=-\dfrac{1}{2}$
	$\Rightarrow y=-\dfrac{1}{2}x+k$. (shown)
4ii	
	Area required = area under the curve – area of triangle $=\int_0^{2k}\sqrt{2k-x}\,dx-\dfrac{1}{2}(2k)(k)$
	$=\int_0^{2k}(2k-x)^{1/2}\,dx-k^2$ $=\left[\dfrac{(2k-x)^{3/2}}{\left(\frac{3}{2}\right)(-1)}\right]_0^{2k}-k^2$
	$=-\dfrac{2}{3}\left[0-(2k)^{3/2}\right]-k^2$ $=\dfrac{2}{3}(2k)^{3/2}-k^2$ units^2
4iii	From (ii), area of D = $\dfrac{4\sqrt{2}}{3}k^{3/2}$ area of $E=k^2$ So, $\dfrac{\text{Area of } D}{\text{Area of } E}=\dfrac{4}{3}$ $\Rightarrow\ \dfrac{\frac{2}{3}(2k)^{3/2}}{k^2}=\dfrac{4}{3}$
	$\Rightarrow\quad (2k)^{3/2}=2k^2$ Squaring both sides,

	$\Rightarrow \quad (2k)^3 = \left(2k^2\right)^2$ $\Rightarrow \quad 8k^3 = 4k^4$ $\Rightarrow \quad k = 2$ (since $k \neq 0$)
5i	when $x = 0$, $y = e^3 - 7$ 
5ii	$y = e^{3-2x} + 7x - 7$ $\frac{\mathrm{d}y}{\mathrm{d}x} = -2e^{3-2x} + 7$
	At point A, gradient is 5 So $-2e^{3-2x} + 7 = 5$ $-2e^{3-2x} = -2$ $e^{3-2x} = 1$ $3 - 2x = 0$
	$x = \frac{3}{2}$ (Shown)
5iii	When $x = \frac{3}{2}$, then $y = e^{3-2(3/2)} + 7\left(\frac{3}{2}\right) - 7 = 1 + \frac{21}{2} - 7 = \frac{9}{2}$ So, A is ($\frac{3}{2}$, $\frac{9}{2}$)
	Equation of normal at A is $\frac{y - \frac{9}{2}}{x - \frac{3}{2}} = -\frac{1}{5}$

	$y-\frac{9}{2}=-\frac{1}{5}x+\frac{3}{10}$ $y=-\frac{1}{5}x+\frac{48}{10}$ $\therefore y=-\frac{1}{5}x+\frac{24}{5}$
5iv	At B, x = 0, so $y=\frac{24}{5}$ So, AB = $\sqrt{\left(\frac{3}{2}\right)^2+\left(\frac{24}{5}-\frac{9}{2}\right)^2}$ $=\sqrt{2.34}$ = 1.529705854 ≈ 1.53 units
6i	Divide the population into 6 strata, namely Public Transport Under 30 years, Public Transport 30-65 years, Public Transport Over 65 years, Privately Owned Vehicle Under 30 years, Privately Owned Vehicle 30-65 years and Privately Owned Vehicle Over 65 years. Calculate the proportionate number to select from each strata.
	No. of Public Transport Under 30 years $=\frac{2000}{9000}\times 200=44.44\approx 44$ No. of Public Transport 30-65 years $=\frac{1500}{9000}\times 200=33.33\approx 33$ No. of Public Transport Over 65 years $=\frac{1000}{9000}\times 200=22.22\approx 22$
	No. of Privately Owned Vehicle Under 30 years $=\frac{400}{9000}\times 200=8.88\approx 9$ No. of Privately Owned Vehicle 30-65 years $=\frac{2500}{9000}\times 200=55.55\approx 56$ No. of Privately Owned Vehicle Over 65 years $=\frac{1600}{9000}\times 200=35.55\approx 36$
	Select the respective number randomly from each strata
6ii	Stratified sampling is more suitable as it is more representative of the employee population.

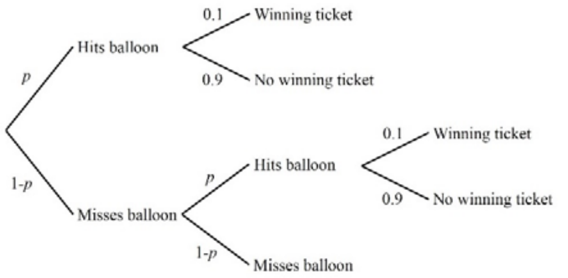
	Or Also systematic sampling may give disproportionately fewer employees from Under 30 years and Private transport as number is very small compared to the other categories.
7i	$\mathrm{P}(A \cup B) = 1 - \mathrm{P}(A' \cap B') = 1 - 0.25 = 0.75$
7ii	$\mathrm{P}(A \mid B) = \mathrm{P}(B \mid A)$ $\frac{\mathrm{P}(A \cap B)}{\mathrm{P}(B)} = \frac{\mathrm{P}(A \cap B)}{\mathrm{P}(A)}$ $\therefore \mathrm{P}(A) = \mathrm{P}(B) \quad (\text{shown})$
7iii	$\mathrm{P}(A \cap B) = 0.5\mathrm{P}(A)$ $\mathrm{P}(A \cap B) = \mathrm{P}(A) + \mathrm{P}(B) - \mathrm{P}(A \cup B)$ $0.5\mathrm{P}(A) = \mathrm{P}(A) + \mathrm{P}(A) - 0.75$ $1.5\mathrm{P}(A) = 0.75$ $\therefore \mathrm{P}(A) = 0.5$ $\therefore \mathrm{P}(A \cap B) = 0.5(0.5) = 0.25$
7iv	$\mathrm{P}(A) \times \mathrm{P}(B) = 0.5 \times 0.5 = 0.25 = \mathrm{P}(A \cap B)$ Since $\mathrm{P}(A \cap B) = \mathrm{P}(A) \times \mathrm{P}(B)$, A and B are independent.
8i	Let S , R and L be the random variable denoting volume of coffee dispensed for a small, regular and large cup respectively. $R - 2S \sim N(250 - 2 \times 240,\ 13^2 + 2^2 \times 15^2)$ $R - 2S \sim N(10, 1069)$
	Required probability = $P(R < 2S)$ = $P(R - 2S < 0)$ = 0.380 (to 3 s.f.)
8ii	Let A be the random variable denoting the total volume of coffee dispensed in a particular hour. $A \sim N(3 \times 120 + 4 \times 250 + 5 \times 350,$ $3 \times 15^2 + 4 \times 13^2 + 5 \times k^2)$ $A \sim N(3110, 1351 + 5k^2)$
	$P(A \geq 3000) > 0.99$

	$P(Z \geq \dfrac{3000-3110}{\sqrt{1351+5k^2}}) > 0.99$ $P(Z \geq \dfrac{-110}{\sqrt{1351+5k^2}}) > 0.99$
	$\dfrac{-110}{\sqrt{1351+5k^2}} < -2.326347877$ $\dfrac{110}{\sqrt{1351+5k^2}} > 2.326347877$ $\dfrac{110}{2.326347977} > \sqrt{1351+5k^2}$ $k^2 < \dfrac{1}{5}\left[\left(\dfrac{110}{2.326347977}\right)^2 - 1351\right]$ $k^2 < 176.963193$ $\lvert k \rvert < 13.30275127$
	$0 < k < 13.3$ (since $k > 0$)
9i	$(\overline{x}, \overline{y})$ = (3.4, 3) 
9ii	From the GC, $r = 0.969$ (to 3sf)
	There is a strong positive linear relationship between the average monthly household income and the average monthly household expenditure. As the average monthly household income increases, the average monthly household expenditure increases proportionately.
9iii	From the GC, equation of the regression line of E on I is, $E = 0.924I - 0.140$ From the GC, equation of the regression line of I on E is,

	$I = 1.02E + 0.352$ $\Rightarrow E = 0.984I - 0.347$
9iv	When $I = 5$, $E = 0.92356(5) - 0.14013 = 4.47767$ The average monthly household expenditure is \$4480 (to 3 s.f.).
	Since $I = 5$ is within the given data range of $0.80 \le I \le 8.35$ (an interpolation), and the value of $r = 0.969$ is close to 1, the predicted value of $E = 4480$ is a reliable estimate.
10	It is assumed that the **outcome of each shot is independent of each other.** **Or** The probability of achieving a successful shot is a constant of 0.48 for all shots.
10i	Let X be the random variable denoting the number of successful shots out of 5. $X \sim \mathrm{B}(5, 0.48)$
	$P(X \ge 3) = 1 - P(X \le 2)$ $= 0.46254$ $= 0.463$ (to 3 s.f.)
10ii	Let Y be the random variable denoting the number of successful shots out of 50. $Y \sim \mathrm{B}(50, 0.48)$ Since $n = 50$ is large, $np = 24 > 5$ and $nq = 26 > 5$,
	$Y \sim \mathrm{N}(24, 12.48)$ approx.
	$P(Y \ge 30) = P(Y > 29.5)$ c.c. $= 0.059749$ $= 0.0597$ (to 3 s.f.)
10iii	$\overline{Y} = \dfrac{Y_1 + Y_2 + \ldots + Y_N}{N}$. Since N is large enough, by CLT, $\overline{Y} \sim \mathrm{N}(24, \dfrac{12.48}{N})$ approx.
	$P(\overline{Y} \ge 23) < 0.998$
	$P(\overline{Y} \le 23) > 0.002$

	$P(Z \le \frac{23-24}{\sqrt{\frac{12.48}{N}}}) > 0.002$ $\frac{23-24}{\sqrt{\frac{12.48}{N}}} > -2.87816$ $N < 103.4$
	<u>Alternative Method</u> Using GC, Y = normcdf(23, 10^{99},24, $\sqrt{\frac{12.48}{N}}$) N $P(\overline{Y} \ge 23)$ 102 0.99787 103 0.99797 104 0.99805 $N \le 103$
	Largest value of $N = 103$
	<u>Alternative Method</u> Let Y be the random variable denoting the number of successful shots out of 50N. $Y \sim \mathrm{B}(50N, 0.48)$ Since require average number of successful shots exceed 23, P(Y > 23N) < 0.998 Using GC, Y = 1- binomcdf(50N, 0.48, 23N) N $P(Y > 23)$ 104 0.99798 105 0.99806 106 0.99815 $N \le 104$ Largest value of N = 104
11i	Unbiased estimate of the population mean

	$= \dfrac{\sum(x-4.5)}{65} + 4.5$ $= \dfrac{21}{65} + 4.5$ $= 4.823076923$ ≈ 4.82 (3 s.f.)
	Unbiased estimate of the population variance $= \dfrac{1}{65-1}(85 - \dfrac{21^2}{65})$ = 1.222115385 ≈ 1.22 (3 s.f)
11ii	$H_0 : \mu = 4.5$ $H_1 : \mu > 4.5$ Level of sig: 1% = 0.01 Test statistic: $z = \dfrac{\bar{x} - \mu_0}{\frac{s}{\sqrt{n}}}$, where $\bar{x} = 4.8231$, $s = \sqrt{1.2221}$ and $n = 65$ Using G.C., p-value = 0.00923 < 0.01
	Therefore we reject H_0, and conclude that there is sufficient evidence at 1% level of significance that the students from the school spend more time reading magazines as compared to the national average.
	No assumption needed. This is because the sample size is large and thus by Central Limit Theorem, $\overline{X}$ follows a normal distribution.
11iii	$H_0 : \mu = \mu_0$ $H_1 : \mu < \mu_0$ The null hypothesis is rejected at 1% level of significance $\Rightarrow \dfrac{4.8231 - \mu_0}{\sqrt{\frac{1.2221}{65}}} < -2.32634$ $\mu_0 > 5.1421$

	$\mu_0 > 5.14$ (to 3 s.f.)
12i	
12ii	$p(0.1)+(1-p)\,p(0.1)=0.05$
	$p(0.1)+(1-p)\,p(0.1)=0.05$ $0.1p+0.1p-0.1\text{p}^2=0.05$ $0.1p^2-0.2p+0.05=0$ $2p^2-4p+1=0$ $p=0.29289$ or $p=1.7071$ $(\text{Reject} \because p>1)$ $\therefore p=0.293$ (shown)
12iii	$\text{P}(\text{winning ticket is from second try} \mid \text{obtains a winning ticket})$ $=\dfrac{\text{P}(\text{winning ticket is from second try} \cap \text{obtains a winning ticket})}{\text{P}(\text{obtains a winning ticket})}$
	$=\dfrac{(1-0.29289)(0.29289)(0.1)}{0.05}$ $=0.41421$ ≈ 0.414 (to 3sf)
12iv	$\text{P}(\text{misses all the balloons in both games})$ $=(1-0.29289)^4$ $=0.25000$ ≈ 0.250 (to 3sf)
12v	Let X be the random variable denoting the number of wins out of n games. $X \sim \text{B}(n\ , 0.05)$

	P(X = 1) = highest probability is the mode.
	Using GC $\begin{array}{c\|l} n & \text{P}(X=1) \\ \hline 18 & 0.37631 \\ 19 & 0.3777353602536 \\ 20 & 0.3777353602535 \\ 21 & 0.37641 \end{array}$
	Since highest probability occurs at X = 19, hence optimal amount is \$19. (\$20 can be also accepted)